

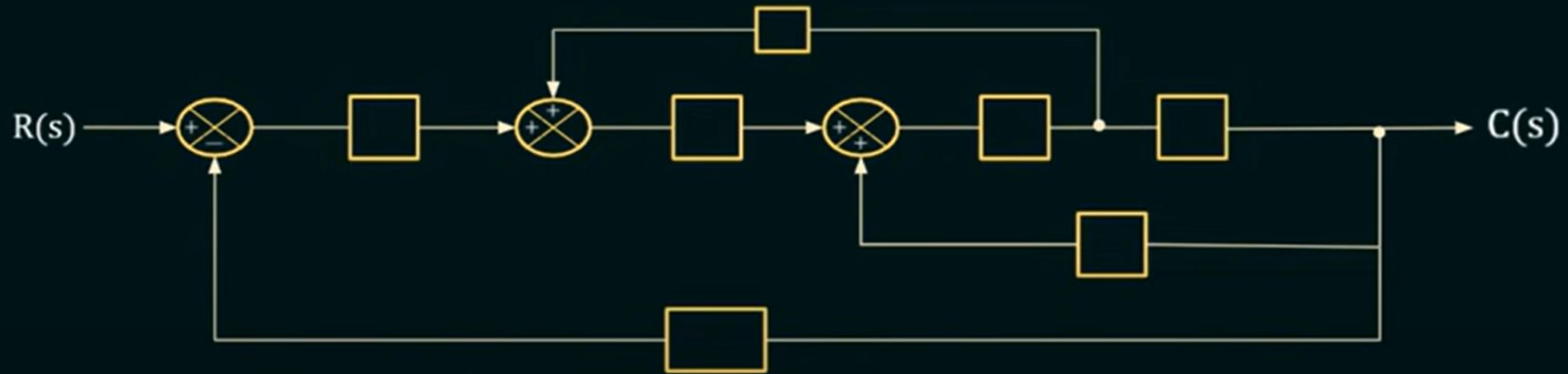
Control System Theory Block Diagram Reduction

<https://youtube.com/playlist?list=PLBlnK6fEyqRineUwP-HPdkQvowXqq5IxA&si=1dXffwOVbx4EpPoL>

Block Diagrams are pictorial representation of a control system.

Block Diagram Representation is used to build a mathematical model of a control system which can be emulated on a computer.

Block diagram Representation is used to calculate the overall transfer function of the system.



Block Diagrams

- ✎ Block Diagrams are pictorial representation of a control system.
- ✎ Block Diagram Representation is used to build a mathematical model of a control system which can be emulated on a computer.
- ✎ Block diagram Representation is used to calculate the overall transfer function of the system.

Elements of a Block Diagram:

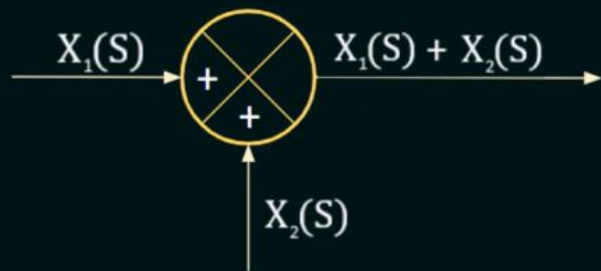


- The signal into the block represents the input.
- The signal out of the block represents the output.
- The block itself represents the Transfer Function of the system.



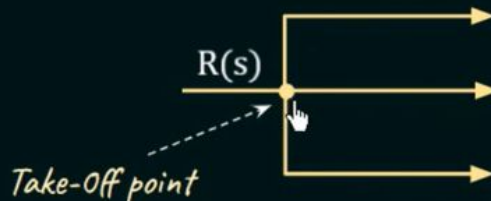
Summing Point:

A Summing Point/Summing Junction in a block diagram represents the dynamic summation of two (or more) signals.



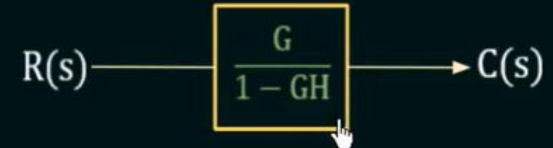
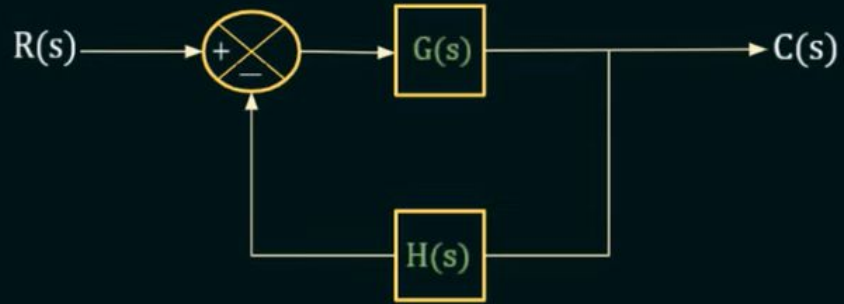
Take-off Point/ Pick-off Point/ Branch point:

Take-off point in a block diagram represents a point where the signal branches out and goes concurrently to the other blocks or summing points.

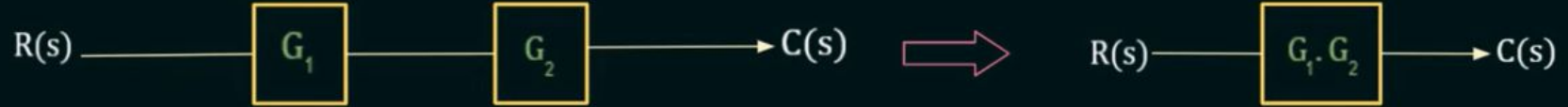


Block Diagram Reduction rule

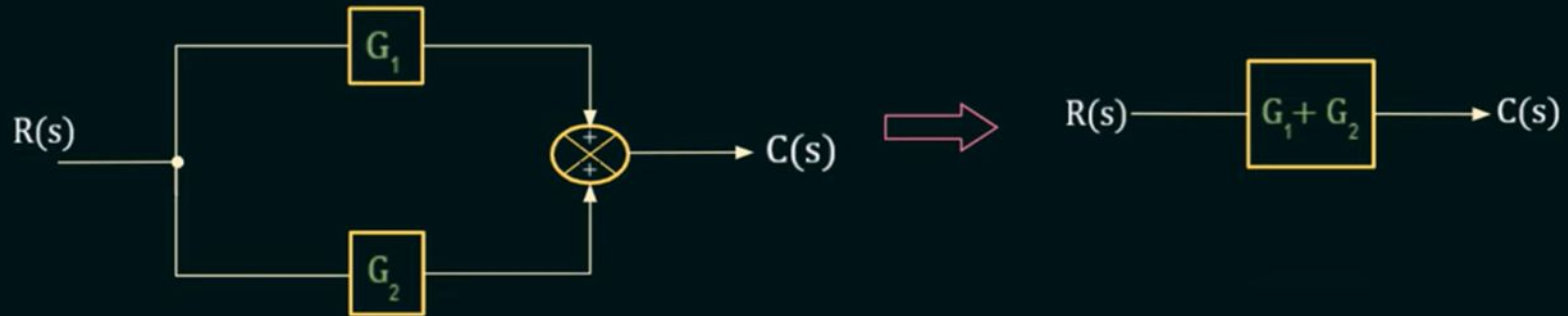
Rule 1: Representation of a closed loop system



Rule 2: When blocks are connected in series/cascade, their gains are multiplied.



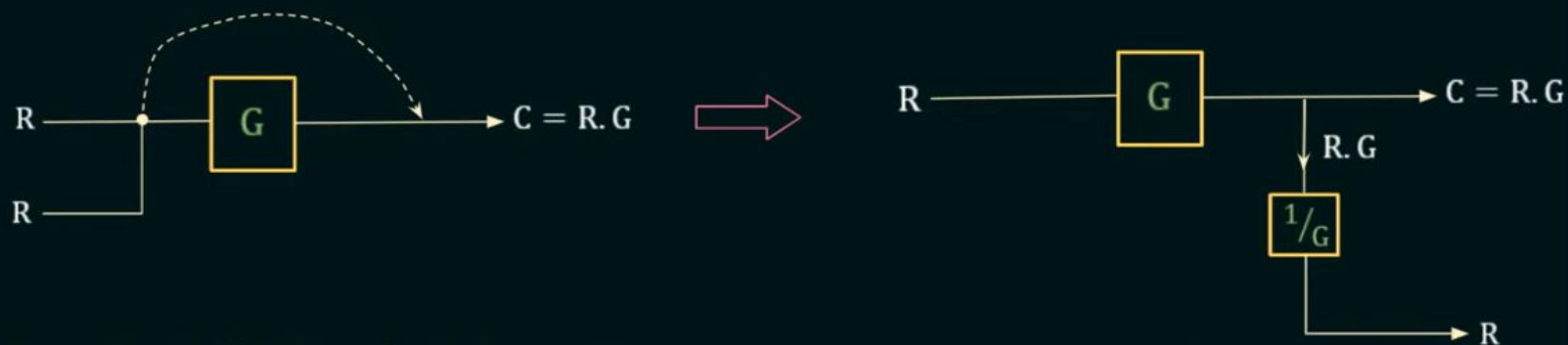
Rule 3: When blocks are connected in parallel, their individual gains are added.



Rule 4: Shifting of take-off point before a block.

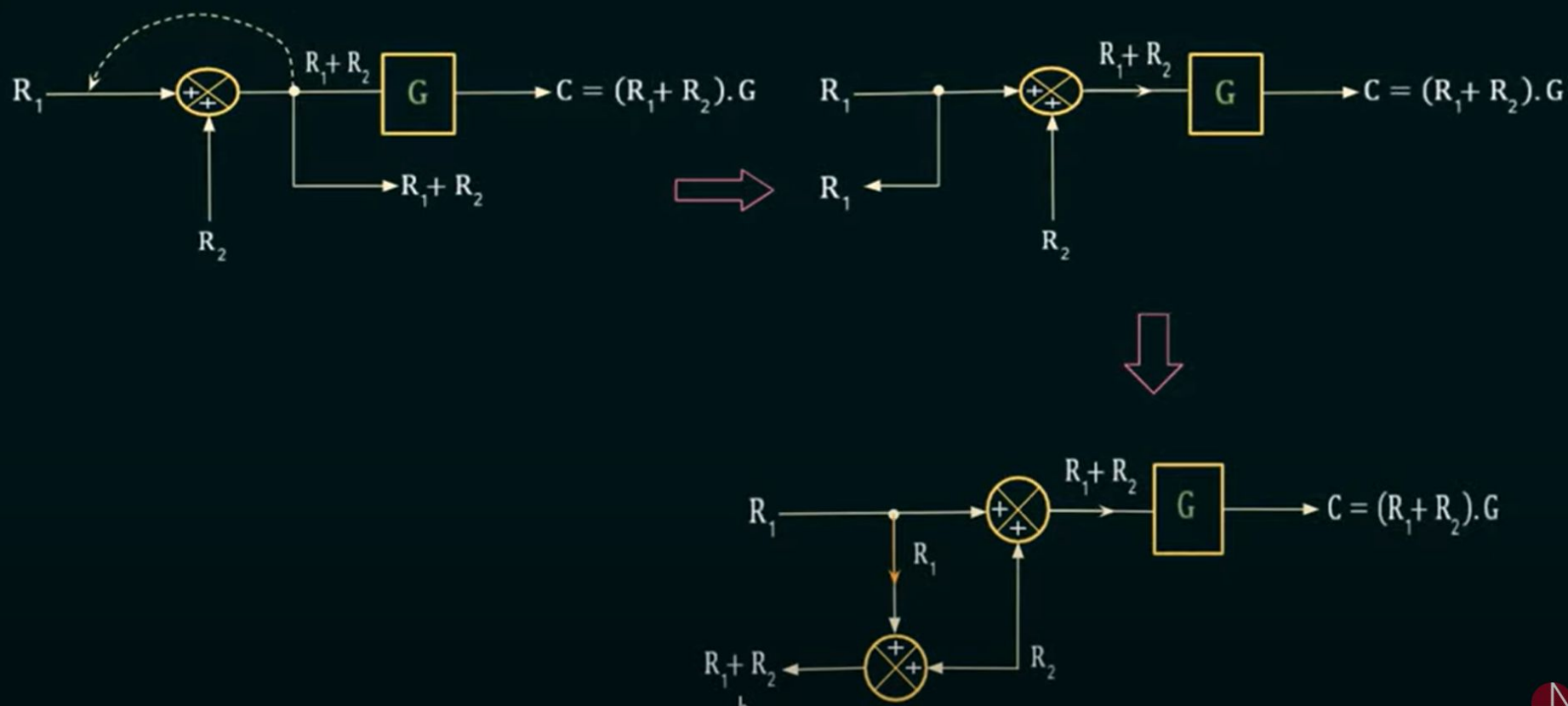


Rule 5: Shifting of take-off point after a block.

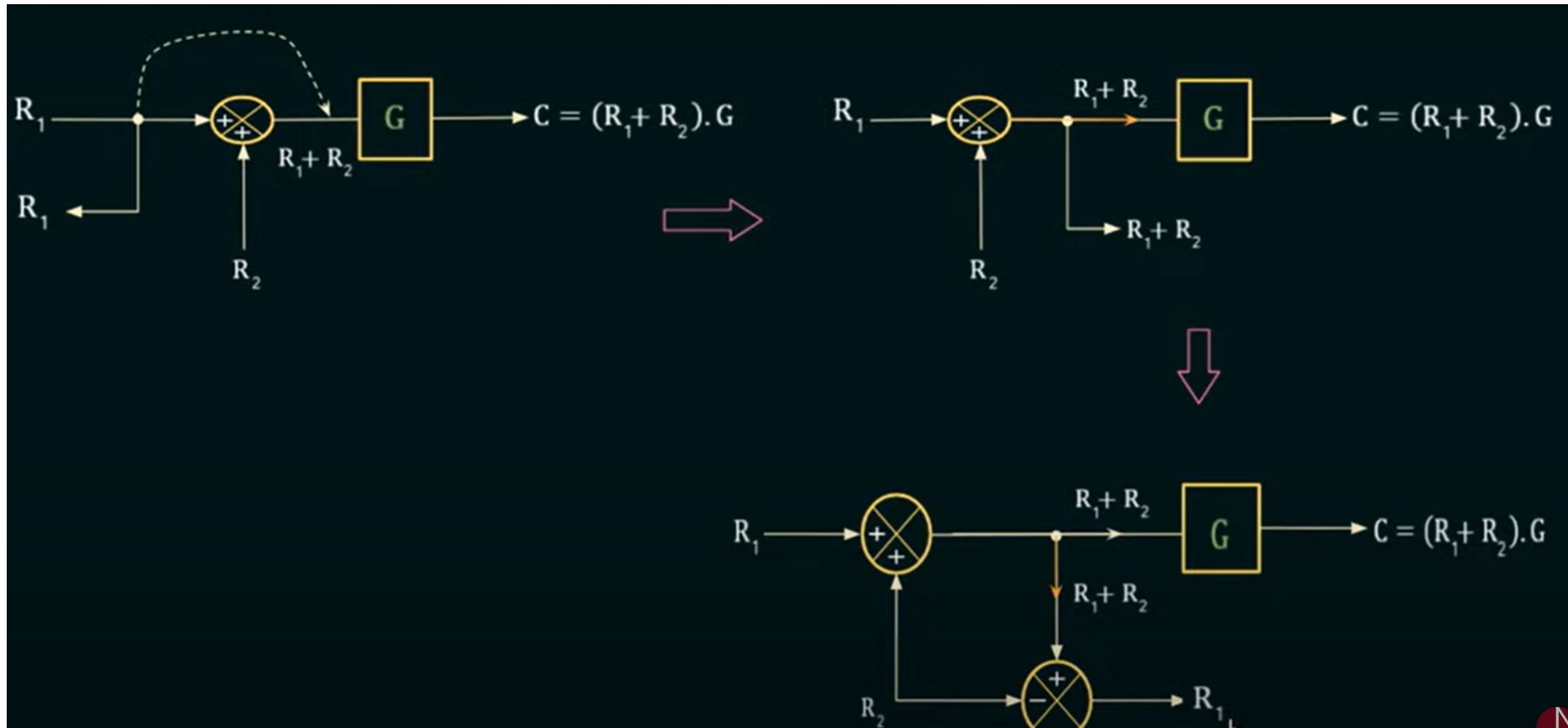


The output of take-off point must be same.

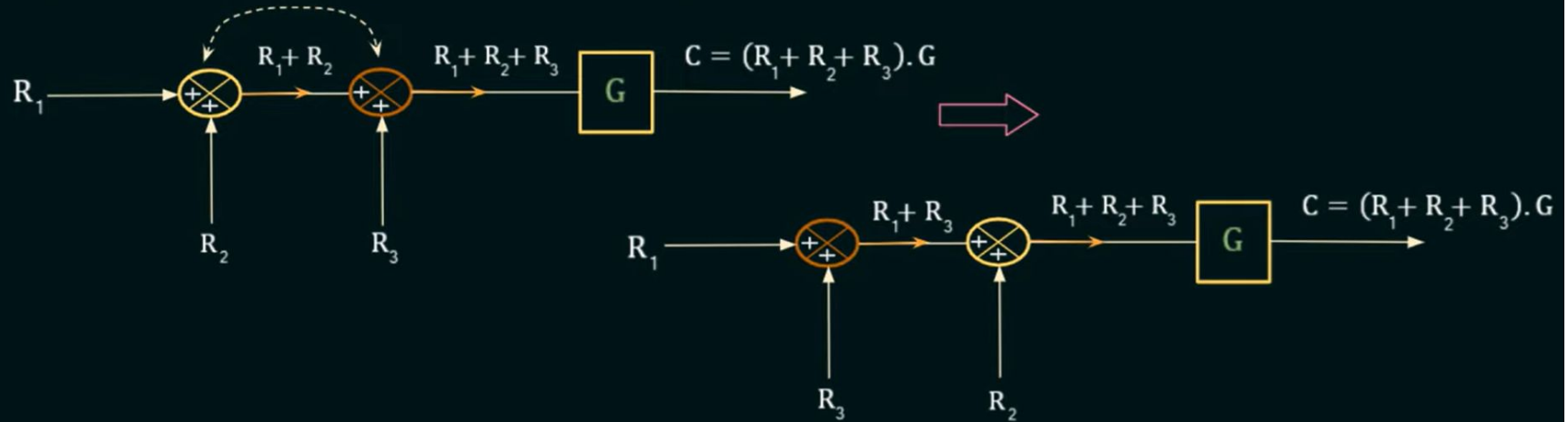
Rule : 6



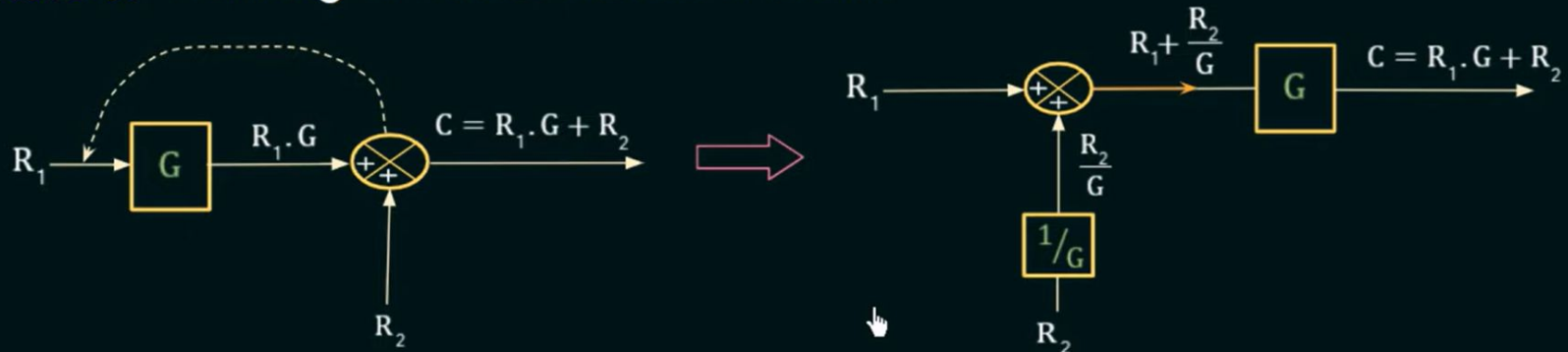
Rule : 7



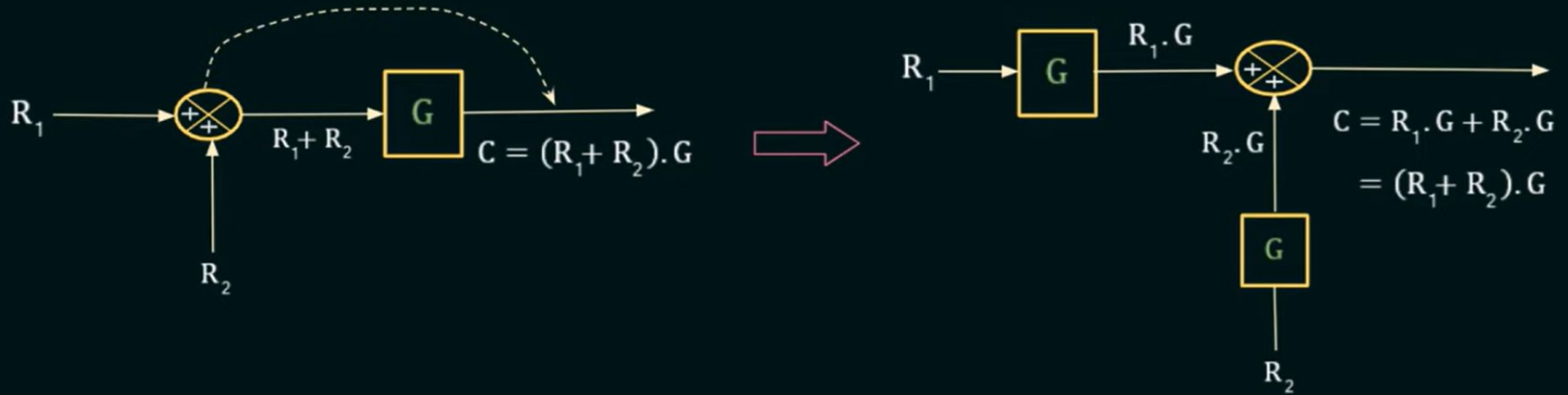
Rule 8: Rearrangement of adders.



Rule 9: Shifting of adder before a block.



Rule 10: Shifting of adder after a block.

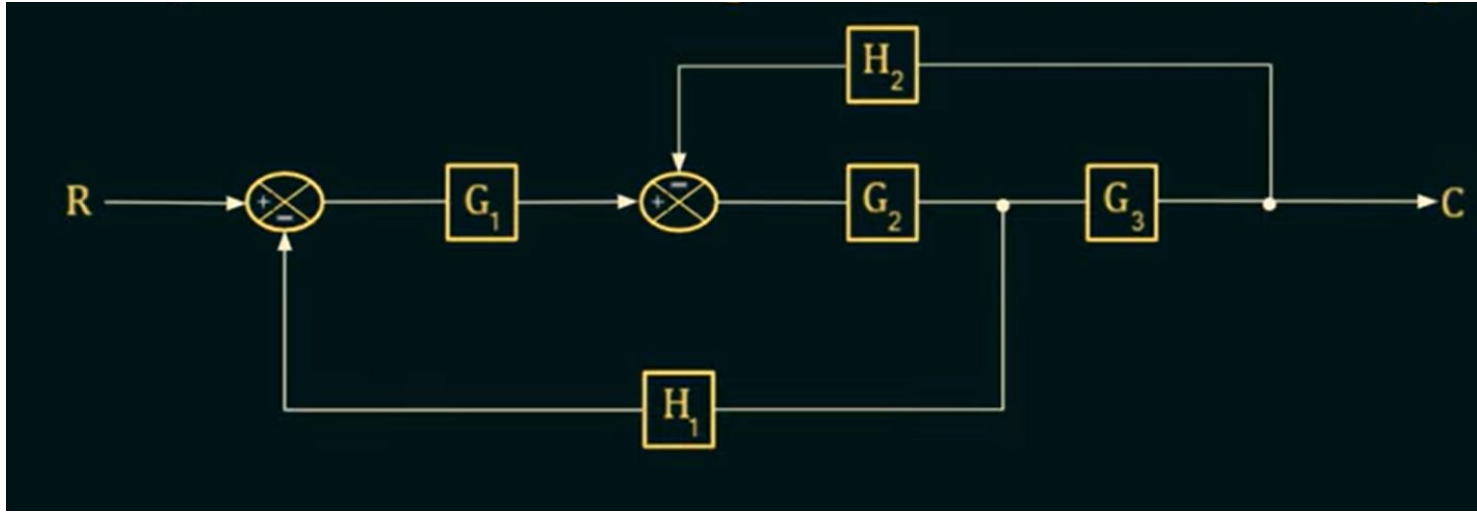


- If we shift the adder before a **block**, then we need to divide the gain of **block** from the input which is shifted.
- If we shift the adder after a **block**, then we need to multiply the gain of the **block** with the input which is shifted.

Block Diagram Reduction Algorithm

- **Rule 1** – Check for the blocks connected in series and simplify.
- **Rule 2** – Check for the blocks connected in parallel and simplify.
- **Rule 3** – Check for the blocks connected in feedback loop and simplify.
- **Rule 4** – If there is difficulty with take-off point while simplifying, shift it towards right.
- **Rule 5** – If there is difficulty with summing point while simplifying, shift it towards left.
- **Rule 6** – Repeat the above steps till you get the simplified form, i.e., single block.

Problem 1:



Rule 1 – Check for the blocks connected in series and simplify.

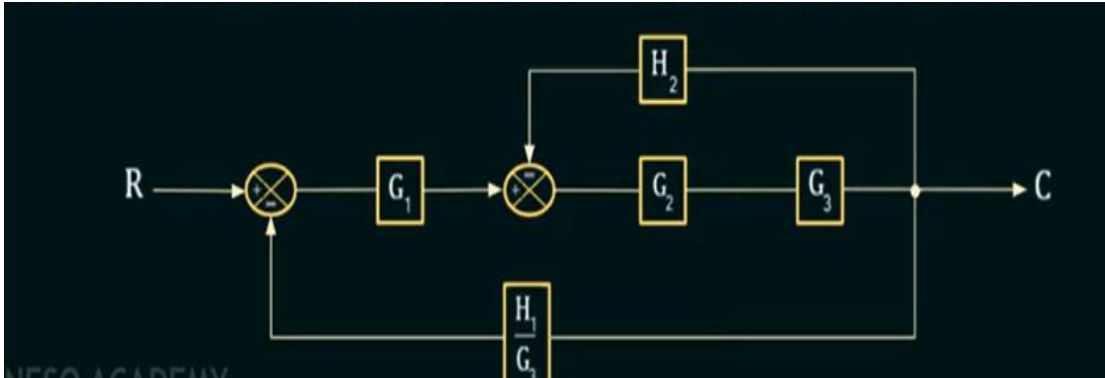
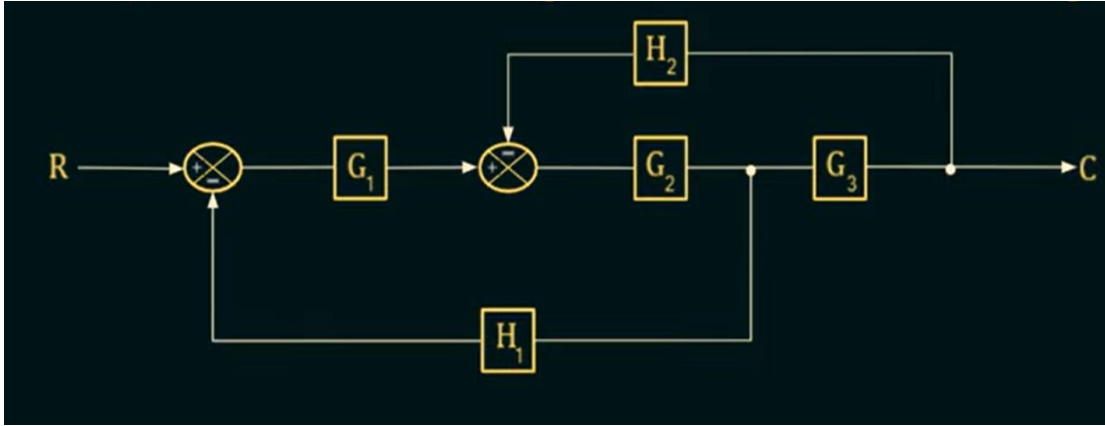
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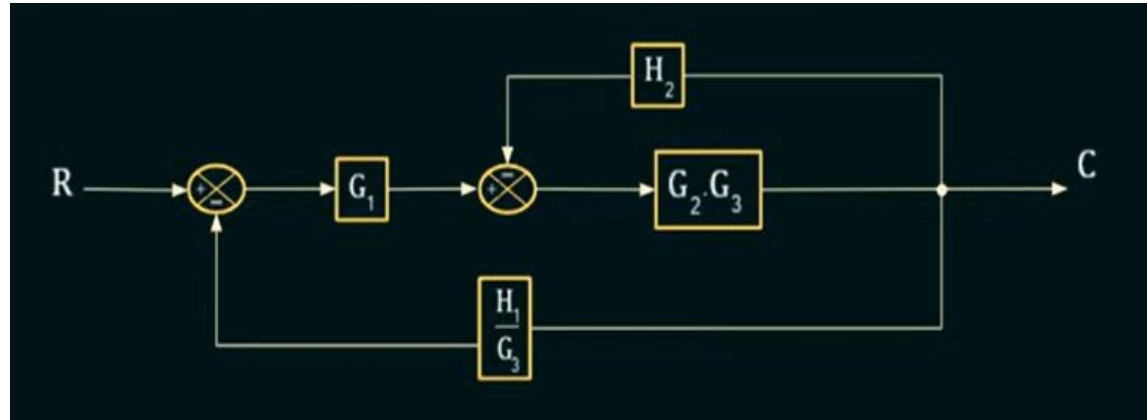
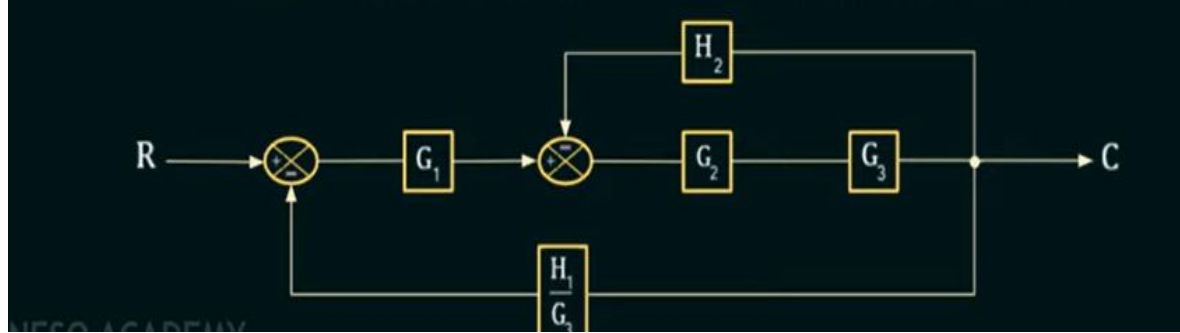
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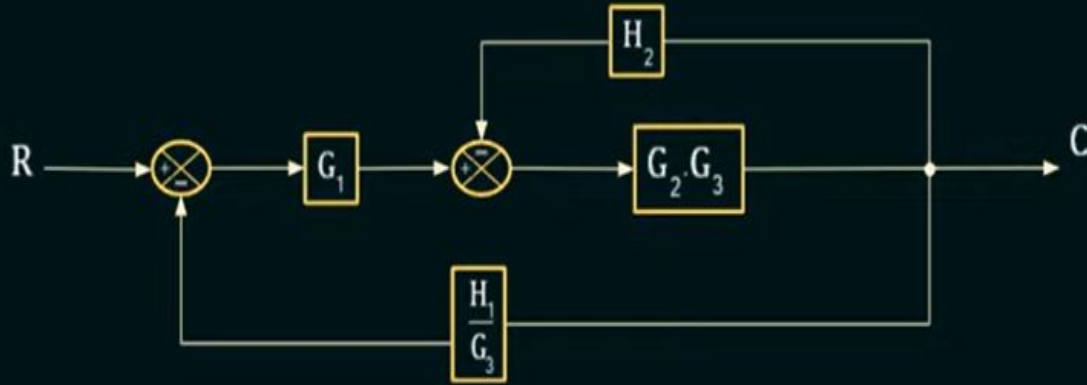
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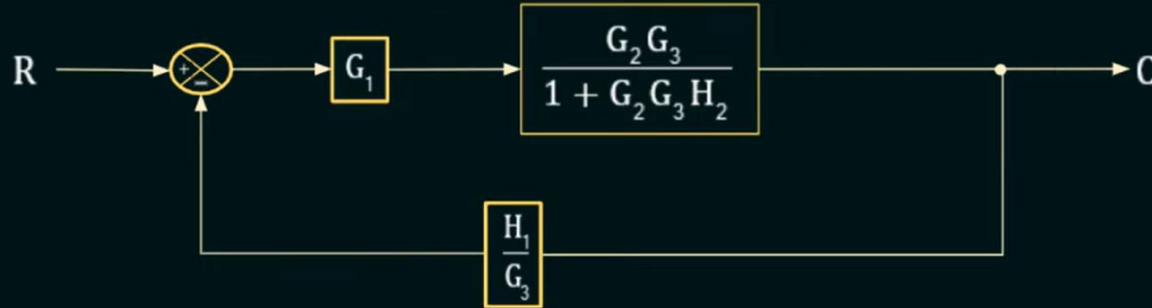
Rule 2 – Check for the blocks connected in parallel and simplify.

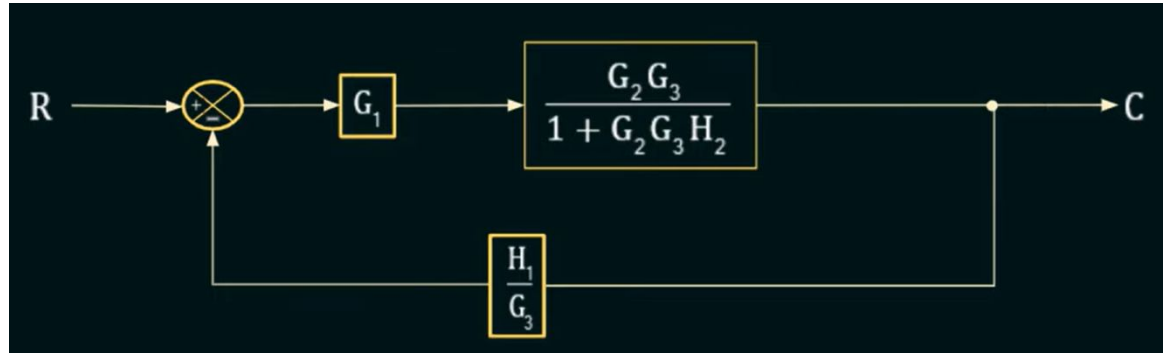
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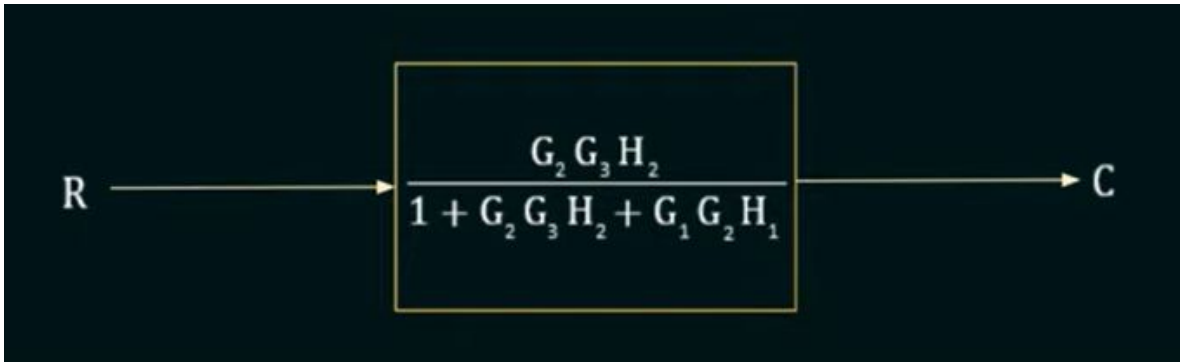
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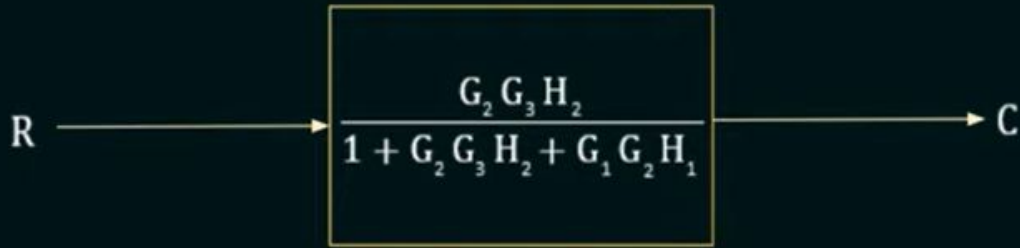
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Result



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Next Class

PID Control

Thank You